

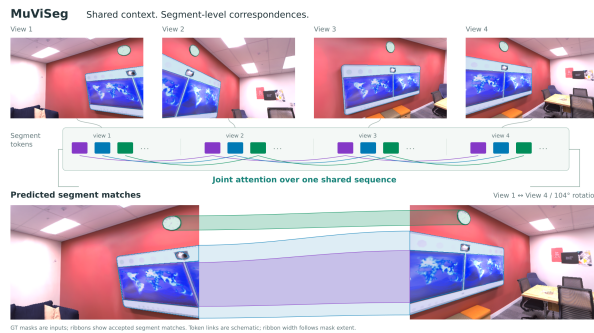
MuViSeg: Multi-View Segment Correspondences from Dense Geometry Priors

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Abstract. Object-level mapping and navigation require correspondences between whole instance segments rather than pixels or keypoints. We study how frozen 3D foundation features should be processed for segment matching and introduce MuViSeg: a learned pairwise attention head on MAST3R, multi-scale VGGT feature fusion, and a joint multi-view head that reasons over segments from several views to recover transitive matches. In zero-shot evaluation on Replica and Virtual KITTI 2 across 0° – 180° viewpoint changes, the MAST3R head improves over a Sinkhorn matcher by 4.85 and 25.9 AUPRC points, respectively. The multi-view head is strongest among our heads at the widest Replica baselines, although Sinkhorn remains competitive. Across 102 HM3D navigation episodes, direct segment matchers and sparse keypoint aggregation are statistically indistinguishable, while RoMa’s dense matches voted into masks score 13.72–17.98 SPL points lower than direct segment matching.

Keywords: Segment matching · 3D foundation models · Multi-view correspondence · Topological navigation



Teaser. Joint multi-view attention recovers segment correspondences across a 104° viewpoint change.

1 Introduction

Image correspondence is a core primitive of visual scene understanding, underpinning object tracking [23], topological navigation [6], and scene-graph con-

struction [7, 12]. But the correspondences these systems ultimately need are not at the level of pixels or sparse keypoints — they are at the level of *objects*. A robot revisiting a room has to decide whether “*this chair seen now is the same chair seen a minute ago*,” and a scene graph is only consistent if the same object carries the same identity across every view it appears in. This is hard for two reasons that compound each other. First, the camera can move arbitrarily between observations, so the same object may be seen from a sharply different angle, scale, or side — the wide-baseline regime where appearance alone is unreliable. Second, the primitive most matchers produce is the wrong one: sparse [3, 15, 24] and dense [5, 29] matchers achieve strong pixel-level accuracy, but a system that needs object identity has to recover it from those low-level matches itself, typically by voting matches inside a segmentation mask — and the resulting associations are only as reliable as that voting rule allows.

A direct route, taken by works like RoboHop [6] and downstream pipelines for topological mapping [6] and 3D scene-graph construction [7], is to move the matching problem itself to the object level: partition each image into instance segments with SAM [9, 23], compute one descriptor per segment, and match by distance-based criteria. Early instantiations relied on appearance-only foundation features [18, 22]; more recent work pools features from *3D foundation models* pre-trained for stereo reconstruction or multi-view structure prediction [11, 30, 31], yielding descriptors that are simultaneously appearance- and geometry-aware and transfer well to wide-baseline matching. SegMASt3R [8] is a representative first step: it takes MASt3R’s frozen patch features, learns a small segment-feature head that maps them to 24-dim per-segment descriptors, and matches them with a parameter-free Sinkhorn [24, 27] optimal-transport layer.

We take this paradigm as our starting point and ask a single design question: *what is the most effective way to process segments for matching?* We answer it by testing three concrete hypotheses, each a different pipeline for turning frozen foundation features into matches. **(H1)** matching capacity is better spent in a learned *cross-segment* head than before the matcher — a LightGlue-style [15] attention head on frozen MASt3R features. **(H2)** preserving multi-scale spatial detail up to the mask boundary, via a DPT-style [21] fusion over VGGT features, yields sharper per-segment descriptors than late pooling. **(H3)**, and central to this work, matching should not be done one pair at a time at all: segments from several views should be scored *jointly*. This last hypothesis is **the method we propose** — a joint multi-view matcher that concatenates segments from N views and reasons over them through shared self-attention with per-view position embeddings, recovering transitive correspondences that strictly pairwise matching cannot reach. We test all three under a stratified zero-shot protocol on Replica [28] (indoor) and Virtual KITTI 2 [2] (outdoor), trained on ScanNet++ [33], with pairs binned by relative camera rotation into four ranges from 0° to 180° . The picture that emerges is regime-dependent: the pairwise cross-segment head is strongest at small-to-moderate baselines and outdoors, while joint multi-view attention is the strongest of our heads at the widest baselines, where pairwise learned heads degrade because two distant views share too few

segments — though the un-learned Sinkhorn baseline remains competitive there. In closed-loop evaluation on HM3D Instance Image Navigation [10,20], dropped into RoboHop [6] without retraining, direct segment-level matchers and sparse SuperPoint+LightGlue aggregation form one cluster, while RoMa’s dense pixel matches voted into masks trail every one of them.

Contributions. (i) A *joint multi-view* segment matcher — our main method — that scores segments from N views together and is the strongest of our heads in the wide-baseline regime, generalising to tuple sizes unseen at training. (ii) A systematic comparison of three segment-processing pipelines that maps out *which* pipeline wins in *which* viewpoint regime, under a stratified zero-shot protocol. (iii) A closed-loop evaluation at scale (102 episodes, 36 scenes) with paired significance testing: direct segment-level matching exceeds the tested RoMa dense-vote baseline by 13.72–17.98 SPL points in the tabulated point estimates, while direct matchers and sparse keypoint aggregation are not statistically separated.

2 Related Work

Sparse and dense feature matching. Sparse pipelines [3,15,24] produce keypoint-level matches; dense methods [4,5,29] predict per-pixel correspondences via 4D correlation volumes or kernelised regression, often on top of frozen self-supervised features [18], with GIM [26] addressing distribution shift through internet-video self-training. LightGlue [15] streamlines the matcher through alternating self- and cross-attention, a DoubleSoftmax scorer, and a learned matchability classifier — a design we lift to the segment level. None of these methods produce object-level associations directly: a downstream system must aggregate sub-object matches inside a mask, and the resulting associations are only as reliable as the voting rule allows.

3D foundation models. DUST3R [31] casts stereo reconstruction as regression of per-pixel pointmaps; MAST3R [11] extends it with prediction heads — the cross-view decoder outputs patch features of shape $(H/16, W/16, 768)$, and a local-feature head additionally yields 24-dim per-patch descriptors for pixel-level matching. MAST3R encodes the two images independently and exchanges information through a CroCo-style [32] cross-view decoder; its features are inherently *pair-dependent*. VGGT [30] takes a contrasting route: a 1B-parameter ViT encoder followed by a 24-layer Aggregator that processes patch tokens from *all* input views in a single shared sequence, with cross-view interaction emerging from self-attention rather than explicit cross-attention. This contrast — pairwise cross-attention vs. joint multi-view self-attention — dictates which heads are architecturally compatible with which backbone, and motivates the two head designs we contrast.

Segment-level matching. RoboHop [6] builds a topological map from SAM masks, pools DINOv2 [18] features over each mask, and matches by cosine similarity

— without explicit multi-view geometry or any learned matching component; TANGO [19] extends this segment-based topological navigation with local metric control, but likewise treats segment association as a fixed feature-similarity step rather than a learned matcher. SegMASt3R [8] is the most directly comparable prior work: it freezes MASt3R’s decoder, takes its 768-dim patch features, and learns a *segment-feature head* (an upsampling MLP) producing 24-dim per-segment descriptors, matched by a parameter-free Sinkhorn [24, 27] solver with a learnable dustbin logit. All learnable capacity sits in the per-segment feature head; the matcher itself does no cross-segment mixing. We instead place capacity in a cross-segment head — a wider projected space with explicit self-/cross-attention — and additionally exploit VGGT’s pair-independent Aggregator to open joint attention across more than two views, a design axis inaccessible to MASt3R-based matchers.

Area-based matching. A parallel line uses area-level correspondence as an intermediate step for point matching: MESA [34, 35] and A2PM [36] establish area matches to narrow the search for keypoints and are ultimately scored by the pose those keypoints yield, so the areas are discarded downstream. MASA [13] associates instances across video frames, where temporal continuity constrains the association. Our setting differs in that segments *are* the output: the topological map stores identities, we score them by instance-ID AUPRC/R@1, and the two views of an object may be separated by a wide baseline rather than a frame interval. The two directions are complementary.

3 Method

We address segment matching with a unified template: a frozen vision backbone Φ produces per-patch features, a class-agnostic segmenter partitions each image into instance masks, masked average pooling produces one descriptor per segment, and a head refines the descriptors and predicts correspondences. Within this template we vary three design axes that follow the openings identified in Section 1: the *placement of learnable capacity* (in the per-segment feature head before a parameter-free solver, as in SegMASt3R [8]; or in a learned cross-segment attention head on top of frozen per-segment descriptors, as in our work), the *choice of backbone* (MASt3R vs. VGGT), and the *breadth of context* (a single pair vs. a tuple of jointly attended views).

Why two backbones. Our object of study is the matching head, yet we deliberately vary the backbone too, because the backbone is not an interchangeable feature extractor here: it determines *which heads are architecturally possible*. MASt3R’s *pair-dependent* features suit a strong pairwise head but cannot be shared across a variable set of views, whereas VGGT’s *pair-independent* features are exactly what a joint multi-view head requires (§3.2). The central axis of this work, single-pair versus joint multi-view matching, is therefore only reachable on VGGT; the backbone comparison is a consequence of studying that axis,

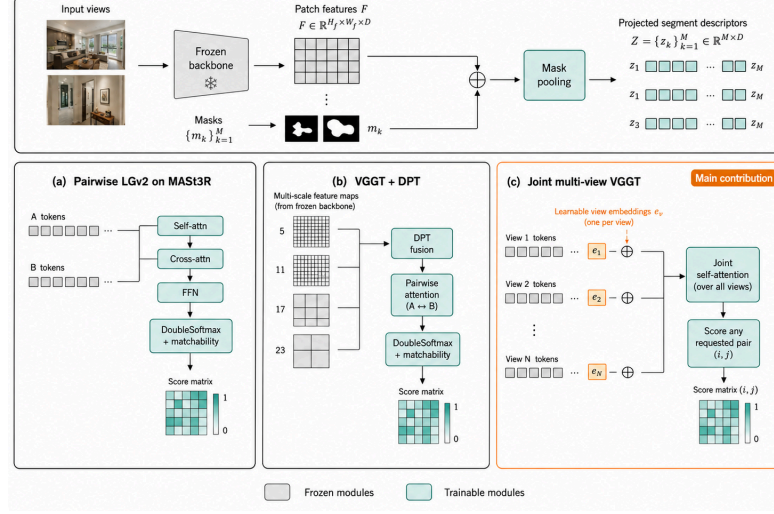


Fig. 1: Architecture of the three matchers (grey: frozen, teal: trainable). **Top:** shared template — frozen backbone $\rightarrow$ per-patch features $\mathbf{F}$, mask-selected and mask-pooled to one descriptor $\mathbf{z}_k$ per segment. The variants differ in feature tap, spatial fusion, and attention context. **(a) LGv2 on MAST3R:** pairwise self-/cross-attention and an FFN, then a DoubleSoftmax scorer with a matchability head. **(b) VGGT+DPT:** a DPT fusion of Aggregator layers $\{5, 11, 17, 23\}$ before pooling; head as in (a). **(c) Joint multi-view VGGT** (main contribution): tokens from all N views, tagged with per-view embeddings $\mathbf{e}_v$, share one joint self-attention, after which the scorer reads off any pair (i, j) .

and we compare head designs under a fixed backbone wherever possible (DPT vs. Joint on VGGT); LGv2 vs. Sinkhorn also holds MAST3R fixed but changes the feature tap as well as the head (Section 3.3). The three models — *SegMASt3R+LGv2*, *SegVGGT-DPT*, and *SegVGGT-DPT Joint* (Fig. 1) — share the scaffolding below.

3.1 Problem formulation

Let $I_A, I_B \in \mathbb{R}^{H \times W \times 3}$ be two views of the same scene. A class-agnostic segmentation model (SAM [9] in our experiments) produces non-overlapping instance masks $\mathcal{M}_A = \{m_1^A, \dots, m_M^A\}$, $\mathcal{M}_B = \{m_1^B, \dots, m_N^B\}$. Mask counts M and N are capped at $M_{\max}$, the maximum number of segments per image. A frozen backbone Φ produces per-patch features $\mathbf{F}_A, \mathbf{F}_B = \Phi(I_A, I_B) \in \mathbb{R}^{H_f \times W_f \times D_{\text{raw}}}$ with raw feature dimension D_{raw} , grid dimensions $(H_f, W_f) = (H/p, W/p)$, and patch size p ($p = 16$ for MAST3R, $p = 14$ for VGGT); $\mathbf{F}_A$ may or may not depend on I_B depending on the backbone (Section 3.2). Per-segment descriptors

are obtained by masked average pooling (masks resized via nearest-neighbour),

$$\mathbf{d}_k^v = \frac{1}{|m_k^v|} \sum_{i \in m_k^v} \mathbf{f}_i^v, \quad (1)$$

where $v \in \{A, B\}$ indexes the view, $\mathbf{f}_i^v$ is the feature at patch i , and $|m_k^v|$ is the number of patches selected by mask m_k^v . A learnable projector $\pi_\theta : \mathbb{R}^{D_{\text{raw}}} \rightarrow \mathbb{R}^D$, with parameters θ and output dimension D , maps them to a shared embedding $\mathbf{z}_k^v = \pi_\theta(\mathbf{d}_k^v)$.

Matching head. The head $g_\theta : (\{\mathbf{z}_k^A\}, \{\mathbf{z}_k^B\}) \mapsto (\mathbf{S}, \{\sigma_k^A\}, \{\sigma_k^B\})$ returns a score matrix $\mathbf{S} \in \mathbb{R}^{M \times N}$ and per-segment matchability logits. Internally, g_θ refines descriptors through self- and cross-attention — letting each segment attend to others in the same image and in the other view — yielding $\tilde{\mathbf{z}}_k^v \in \mathbb{R}^D$. Let $\tau > 0$ denote the softmax temperature. From the scaled affinity $A_{ij} = \tilde{\mathbf{z}}_i^A \cdot \tilde{\mathbf{z}}_j^B / (\tau \sqrt{D})$ we form two conditional distributions: a row-softmax normalising each row of $\mathbf{A}$ over the columns, and a column-softmax normalising each column over the rows,

$$p(j | i) = \frac{\exp A_{ij}}{\sum_{j'} \exp A_{ij'}}, \quad p(i | j) = \frac{\exp A_{ij}}{\sum_{i'} \exp A_{i'j}}. \quad (2)$$

The *DoubleSoftmax* score is the log of their product,

$$S_{ij} = \log p(j | i) + \log p(i | j), \quad (3)$$

which is large only when segment j is the best match for i and i is the best match for j — i.e. it rewards mutually most-similar pairs.

Matchability and the dustbin. Not every segment has a counterpart in the other view — some are occluded, out of frame, or simply absent — so the matcher must be able to abstain. Each segment carries a *matchability logit* σ_k^v , predicted by a small classifier on top of $\tilde{\mathbf{z}}_k^v$, estimating the probability that this segment has any correspondence in the other view. Following SuperGlue [24], $\mathbf{S}$ is also augmented with a learnable *dustbin* row and column acting as an explicit “no-match” slot. At inference, segment i in A is matched to $j^* = \arg \max_j S_{ij}$ iff $\text{sigmoid}(\sigma_i^A) > 0.5$ and S_{ij^*} exceeds the dustbin score; otherwise it is left unmatched.

3.2 Backbones: MAST3R and VGGT

MASt3R [11] processes I_A, I_B through a ViT-Large encoder independently, then a CroCo-style decoder [32] alternates self- and cross-attention between the two streams, producing patch features of shape $(H/16, W/16, 768)$; a pre-trained local-feature head additionally maps them to 24-dim per-patch descriptors. We take this head’s output, frozen, as our backbone features and pool at patch resolution. This differs from SegMASt3R, which keeps only the 768-dim patch features frozen and learns its own segment-feature MLP on top. Either way, features are inherently *pair-dependent*: $\mathbf{F}_A$ cannot be computed without I_B .

VGGT [30] uses a DINOv2-based ViT encoder followed by a 24-layer Aggregator that processes patch tokens from all input views in a single shared sequence (token dim 2048, patch size 14); cross-view interaction emerges from self-attention. Features are *pair-independent* — they do not change when other views are added to or removed from the same forward pass — which is what makes per-image precomputation and multi-view extensions feasible. This contrast (pairwise cross-attention vs. joint multi-view self-attention) dictates which heads are architecturally compatible with which backbone.

3.3 Three matching heads

SegMASt3R+LGv2 (Fig. 1a). A two-layer MLP lifts the frozen 24-dim MAST3R descriptors into the shared 128-dim space, $\mathbf{z}_k = \text{Linear}_{64 \rightarrow 128} \circ \text{LN} \circ \text{GELU} \circ \text{Linear}_{24 \rightarrow 64}(\mathbf{d}_k)$. Three transformer blocks then refine the two descriptor sets, each block applying weight-shared self-attention on each view, cross-attention in both directions (with separate LayerNorms on queries and keys/values), and a position-wise FFN with expansion factor 4. A two-layer classifier $\text{Linear}(128 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 1)$ predicts the per-segment matchability logits. The full head adds $\sim 800\text{K}$ trainable parameters on top of MAST3R’s $\sim 1\text{B}$ frozen weights. As noted above, LGv2 and Sinkhorn differ in both feature tap and matching head, so their comparison is not head-only.

SegVGGT-DPT (Fig. 1b). A minimal VGGT-based reference, *SegVGGT* (*single-layer*), pools features from the last Aggregator layer and feeds them through a small projector to 128 dim plus the same attention stack as LGv2. Pair-independent Aggregator features make this variant precomputable per-image. Single-layer pooling, however, is spatially coarse (patch size 14 gives a 24×36 map for 336×512 input); the deeper layers of a ViT backbone tend to be semantically rich but spatially coarse, while shallower layers retain finer spatial structure — a well-known layer-wise trade-off [21]. We adopt the DPT [21] fusion strategy to recover boundary precision: features from Aggregator layers $\{5, 11, 17, 23\}$ are reshaped into spatial grids, projected to 256 channels by 1×1 convolutions, and fused from deepest to shallowest through Residual Convolutional Units (pipeline notation in Supplementary Section B). The fused map is mask-pooled (Eq. 1) and projected to 128 dim; the remaining attention stack and matchability head are identical to LGv2. Since fusion happens *before* pooling, per-image precomputation is no longer possible. Total trainable parameters: $\sim 10.7\text{M}$.

SegVGGT-DPT Joint (Fig. 1c). The pairwise heads above treat each pair in isolation; when viewpoint change is large, the two images may share very few segments, and intermediate views often bridge the gap. The Joint variant exploits this by sharing attention across N views at once. Backbone, DPT fusion, pooling, and projection are applied independently to each view; the resulting per-view descriptor sets $\mathbf{z}^{(v)}$ are then concatenated into a single sequence $\mathbf{x} = [\mathbf{z}^{(1)} + \mathbf{e}_1; \dots; \mathbf{z}^{(N)} + \mathbf{e}_N]$ with learnable per-view position embeddings

$\mathbf{e}_v$ added once before the first layer. Three joint attention layers refine the sequence (self-attention + FFN, no cross-attention — a shared sequence already lets any segment attend to any other across views); the same DoubleSoftmax scorer is then applied to any requested pair. We train with $N = 4$ and evaluate at $N \in \{2, 4, 6, 8\}$ without retraining; because joint attention treats views symmetrically and the position embeddings are sparse (only indices 0–3 are exercised during training), the model generalises across N without collapse.

Training and hyperparameters. We set $D = 128$, $M_{\max} = 100$ segments per image, $\tau \equiv 1$ for the MAST3R head, and a learned $\tau \leq 1$ for the VGGT heads. All models are trained on ScanNet++ [33] with ground-truth correspondences from projected 3D instance annotations ($G_{ij} = \mathbb{K}[\text{id}(m_i^A) = \text{id}(m_j^B)]$, where $\text{id}(m)$ denotes the ground-truth instance ID of mask m and $\mathbb{K}[\cdot]$ equals one when its condition is true and zero otherwise). The loss is a normalised NLL over ground-truth pairs $\mathcal{P}$, $\mathcal{L}_{\text{match}} = -\frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} S_{ij}$, plus binary cross-entropy matchability with weight $\lambda = 0.3$. For the Joint model the loss is normalised per-tuple. AdamW [17], learning rate 10^{-4} , weight decay 10^{-4} , 500 warmup steps, cosine schedule, bf16, $2 \times$ NVIDIA RTX 5090 with DDP. Supplementary Section C collects the documented implementation settings.

4 Experiments

We evaluate all three matchers under a single stratified protocol, compare direct segment matching with sparse and dense aggregation in closed-loop HM3D navigation (Section 4.5), and analyse a multi-view sweep over N (Section 4.4). The full temperature-clamp ablation is in Supplementary Section A.

4.1 Setup

Benchmarks. All matchers are trained only on ScanNet++ (Section 3) and evaluated *zero-shot* — never exposed to the test domains during training — on **Replica** [28] (indoor) and **Virtual KITTI 2** [2] (outdoor driving), which together probe indoor coverage, outdoor transfer, and a large appearance gap. Pairs are stratified into four angular bins by relative camera rotation (0–45°, 45–90°, 90–135°, 135–180°); ground-truth correspondences match segments sharing an instance ID. Replica and Virtual KITTI 2 are both rendered datasets; we therefore additionally report ScanNet++ validation (Table 3) on real captured imagery from held-out scenes, which is indoor and in-domain rather than zero-shot.

Baselines. We compare against five baselines: **SegMASt3R (Sinkhorn)** [8] (our most direct prior work; frozen MAST3R patch features $\rightarrow$ learnable segment-feature MLP $\rightarrow$ Sinkhorn solver with learnable dustbin); **vizEnc+DINOv2** [18] (the original RoboHop [6] design — here *vizEnc*, short for *visual encoder*, denotes a frozen appearance-only backbone whose per-mask features are pooled

and matched by cosine similarity); **vizEnc+Radio** (the same recipe with frozen AM-RADIO [22] features in place of DINOv2); **LiftFeat** [16] (a recent 3D-geometry-aware sparse local feature matcher, lifted to segment level by aggregating keypoint descriptors within each mask); and **DA3 (Sequential)** [14] (Depth Anything 3 features pooled over masks; DA3 has no native matcher, so correspondences are obtained by greedy sequential assignment under cosine similarity until the next-best candidate’s matchability falls below a threshold). We also report *SegVGGT (single-layer)*, a VGGT-based head without DPT fusion, as an ablation reference: on ScanNet++ validation it reaches 0.858 AUPRC versus 0.886 for SegVGGT-DPT, isolating the contribution of multi-scale fusion to +2.8 AUPRC points.

Metrics. Our headline metric is **AUPRC** (area under the precision–recall curve), a threshold-free summary that rewards ranking true correspondences above false ones. We complement it with **R@1** and **R@5** (recall at 1/5): the fraction of query segments whose ground-truth match is the top-ranked candidate, or within the top five. All are reported per angular bin and as a query-weighted average.

4.2 Indoor segment matching: Replica

Table 1: Segment matching on Replica (indoor, 3,200 pairs, 28,022 queries), stratified by relative camera rotation. AUPRC reported per angular bin; overall AUPRC, R@1 and R@5 are query-weighted across bins. Best per column in **bold**, second best underlined.

Model	AUPRC by bin				Overall	Overall	Overall
	0–45°	45–90°	90–135°	135–180°	AUPRC	R@1	R@5
vizEnc+Radio [22]	41.91	38.99	41.25	38.36	40.69	58.20	79.04
LiftFeat [16]	47.17	32.13	25.19	24.92	35.69	40.21	67.57
vizEnc+DINOv2 [18]	50.78	47.24	49.49	50.15	49.48	59.29	80.29
DA3 (Sequential) [14]	69.36	55.91	38.99	33.23	54.29	73.96	73.96
SegMASt3R (Sinkhorn) [8]	80.48	77.04	<u>79.83</u>	81.72	79.63	81.45	94.21
SegVGGT-DPT (ours)	81.20	80.93	78.07	73.36	79.70	74.10	90.11
SegVGGT-DPT Joint, $N=4$ (ours)	<u>81.66</u>	<u>83.03</u>	81.85	<u>77.69</u>	<u>81.37</u>	75.64	90.42
SegMASt3R+LGv2 (ours)	91.17	83.59	77.09	70.28	84.48	<u>77.46</u>	<u>93.63</u>

Table 1 reports Replica. SegMASt3R+LGv2 attains 84.48% overall AUPRC, +4.85 over the Sinkhorn baseline, concentrated at narrow baselines (+10.7 at 0–45°); but at 135–180° it drops 11.4 points *below* the baseline. This reversal is informative: the learned head sharpens the regime well-represented in training yet extrapolates worse than the parameter-free matcher to low-overlap, high-rotation pairs. There, SegVGGT-DPT Joint ($N=4$) takes over, leading at 90–135° and 135–180° and beating the pairwise DPT variant by 4.3 AUPRC at the widest bin — isolating the multi-view contribution. The result is a regime

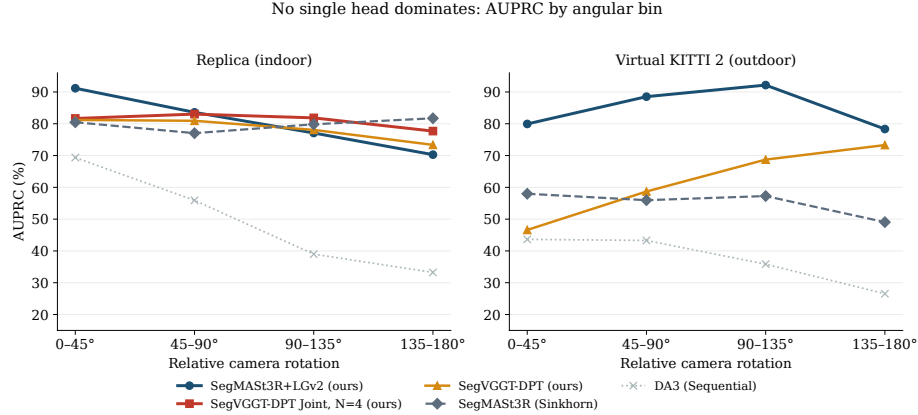


Fig. 2: AUPRC per relative-rotation bin on Replica (left) and Virtual KITTI 2 (right). On Replica, LGv2 leads at narrow baselines but is overtaken by Joint (and even the Sinkhorn baseline) past 90°; on VKITTI2 LGv2 dominates every bin. The crossover is the central observation: the best head depends on the viewpoint regime.

split, not a single winner (Fig. 2): LGv2 leads overall and at narrow baselines, while Joint leads among our heads at the two widest bins. All three are at or above the Sinkhorn baseline overall; DPT’s 0.07-point margin is not evidence of a meaningful separation.

4.3 Outdoor zero-shot: Virtual KITTI 2

Table 2: Segment matching on Virtual KITTI 2 (outdoor driving, 3,200 pairs, 10,308 queries), under the same stratified protocol as Replica. All models trained on ScanNet++ and evaluated zero-shot. Best per column in **bold**, second best underlined.

Model	AUPRC by bin				Overall	Overall	Overall
	0–45°	45–90°	90–135°	135–180°	AUPRC	R@1	R@5
vizEnc+Radio [22]	29.01	28.29	23.47	21.12	27.54	43.38	66.89
vizEnc+DINOV2 [18]	31.86	31.05	25.50	23.40	30.22	46.68	70.34
LiftFeat [16]	35.40	30.00	27.21	20.88	32.35	47.21	68.04
DA3 (Sequential) [14]	43.64	43.29	35.84	26.53	40.67	53.74	75.40
SegMASt3R (Sinkhorn) [8]	<u>57.98</u>	55.96	57.26	49.06	<u>56.91</u>	<u>63.67</u>	<u>81.57</u>
SegVGGT-DPT (ours)	46.60	<u>58.69</u>	<u>68.72</u>	73.29	52.28	42.80	67.27
SegVGGT-DPT Joint, $N=4$ (ours)	47.50	57.54	<u>70.10</u>	<u>76.92</u>	52.85	42.93	66.75
SegMASt3R+LGv2 (ours)	79.94	88.52	92.16	78.34	82.78	73.25	96.22

LGv2 reaches 82.78% AUPRC overall on Virtual KITTI 2 (Table 2), +25.87 over the Sinkhorn baseline and +30.50 over pairwise VGGT. MAST3R’s explicit

cross-view geometric coupling transfers from indoor training to outdoor driving better than VGGT’s joint self-attention does, and the LGv2 head amplifies this signal rather than diluting it. Both VGGT heads exhibit an inverted driving-specific angular pattern: SegVGGT-DPT rises from 46.60% to 73.29% from the narrowest to the widest bin, consistent with difficulty separating near-identical segments at narrow baselines; LGv2 removes this inversion. Consistent with the indoor Replica trend, the joint multi-view head helps most exactly where the baseline is widest: SegVGGT-DPT Joint ($N=4$) improves the pairwise VGGT variant from 73.29 to 76.92 at 135–180°, the single bin where sharing segments across views matters most, even though its overall AUPRC is essentially tied with the pairwise variant (52.85 vs. 52.28).

Table 3: ScanNet++ validation on real captured imagery from held-out scenes. The evaluation is indoor and in-domain, not zero-shot. AUPRC, R@1, and R@5 are percentages.

Matcher	AUPRC	R@1	R@5
SegMASt3R (Sinkhorn)	88.9	83.3	97.1
SegVGGT-DPT (ours)	89.6	82.2	97.6
SegVGGT-DPT Joint (ours)	90.6	85.5	97.1
SegMASt3R+LGv2 (ours)	91.0	85.4	98.5

4.4 Multi-view sweep: effect of N

Figure 3 plots the Joint model at $N \in \{2, 4, 6, 8\}$ per angular bin (trained with $N=4$, evaluated at each N without retraining). On Replica N has almost no effect (81.10–81.57 overall AUPRC): indoor scenes have dense per-view overlap, so extra views add little. On Virtual KITTI 2, N helps specifically at wide baselines: the 135–180° bin improves from 71.05 ($N=2$) to 78.22 ($N=6$) before saturating, while narrow bins stay flat around 47.5% — a per-pair VGGT bottleneck that adding views cannot solve. The model extrapolates to $N = 6, 8$ without collapse even though the position-embedding indices 4–7 are never directly supervised, consistent with joint attention learning view-relative rather than view-absolute patterns.

Temperature calibration. Clamping the learned DoubleSoftmax temperature to $\tau \leq 1$ cuts the dominant *false-dustbin* errors by 36% relative, re-allocating the error budget toward harder *wrong-match/false-match* cases; for this head, calibrated scoring rather than richer descriptors is the binding constraint. The full error-budget decomposition is in Supplementary Section A.

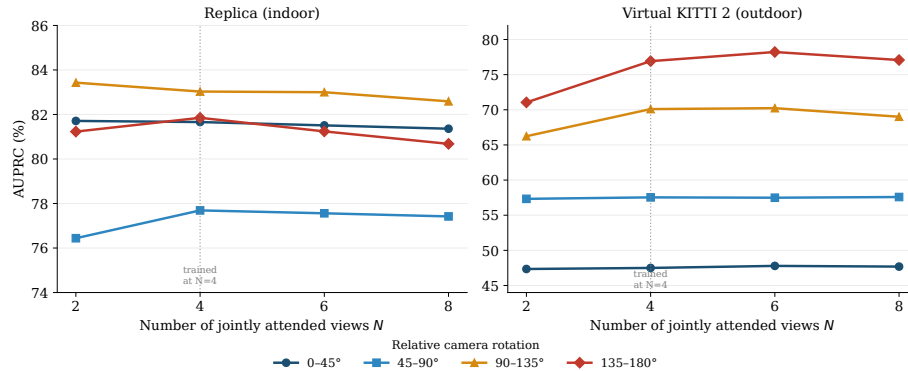


Fig. 3: Effect of the number of jointly attended views N on SegVGGT-DPT Joint (trained at $N=4$, evaluated at $N \in \{2, 4, 6, 8\}$ without retraining). Replica (left) is flat in N (dense overlap); on VKITTI2 (right) extra views help only at the widest baseline (71.1→78.2 AUPRC at 135–180°) and saturate by $N=6$.

4.5 Closed-loop downstream: HM3D Instance Image Navigation

Each matcher replaces the RoboHop [6] online localizer for HM3D Instance Image Navigation [10, 20], without retraining. The agent receives a reference image of a single object instance and must reach within 1 m geodesic distance (Habitat-Sim [25], 250-step cap). RoboHop’s map is a topological graph of SAM segments built from a teach run; at inference, FastSAM [37] segments the current observation, the online matcher compares query segments against a ± 8 -frame window of the map, and a Dijkstra planner returns the next goal segment. We retain the released LightGlue-built teach-run map, isolating query-time matching as in SegMASt3R [8]. Evaluation is on the full HM3D v2 *val* split — 102 episodes across 36 scenes. Baselines are LightGlue [15] (RoboHop default, SuperPoint+LightGlue aggregated to segments) and the released SegMASt3R (Sinkhorn) checkpoint [8]. We evaluate all three of our matchers, with Joint at $N=2$ and $N=4$, an oracle GT metric localizer as an upper bound, and RoMa [5] → segment vote as a dense baseline aggregating pixel correspondences inside masks. Metrics [1]: Success Rate (SR), Success-weighted Path Length (SPL), Soft SPL.

Comparisons are paired: every matcher sees the same episode list with identical localizer settings, and only the online matcher differs. We test success with an exact McNemar test and per-episode SPL with a paired bootstrap (10k resamples). For each endpoint separately, we apply Holm correction across all $\binom{8}{2} = 28$ pairwise method comparisons, including the oracle.¹

Two separations survive correction: the oracle outperforms every matcher, and RoMa voted into masks falls below every direct segment-level matcher

¹ The seed is fixed at 42. On a separate 10-episode subset, repeated runs varied by roughly ± 6 SPL points due to non-deterministic GPU kernels; this diagnostic is not the 102-episode significance test.

Table 4: Closed-loop HM3D v2 *val* (102 episodes, 36 scenes). Matchers share the episode list, map, masks, and localizer settings; the oracle uses ground-truth metric localization. Values are percentages, grouped by method rather than performance. The five direct segment-level configurations and sparse SuperPoint+LightGlue baseline are not statistically separated after Holm correction; no ranking is claimed.

Online matcher	SR	SPL	Soft SPL
Oracle (GT metric localizer)	78.43	78.33	84.29
SuperPoint+LightGlue $\rightarrow$ segment vote [15]	44.12	44.10	53.16
SegMASt3R (Sinkhorn) [8]	42.16	42.36	52.21
SegMASt3R+LGv2 (ours)	41.18	41.17	51.89
SegVGGT-DPT (pairwise, ours)	42.16	42.05	50.17
SegVGGT-DPT Joint ($N=2$, ours)	45.54	45.43	53.60
SegVGGT-DPT Joint ($N=4$, ours)	43.14	43.03	53.45
RoMa $\rightarrow$ segment vote [5]	27.45	27.45	37.40

($p = 0.001$ – 0.008). Differences in Table 4’s SPL point estimates span 13.72–17.98 points; this range is not a bootstrap interval. The five direct segment configurations and sparse keypoint aggregation form one cluster ($p = 0.442$ – 0.860 , with SPL intervals covering zero), so no ranking is claimed. The separation is specific to the tested *dense* pixel-vote baseline: RoMa underperforms, whereas SuperPoint and LightGlue aggregated to segments remain in the cluster. This is not evidence against sparse aggregation or dense matchers in general.

4.6 Discussion

Two factors explain the offline regime split. First, the backbone gates the head: MAST3R’s pair-dependent decoder feeds the strongest pairwise head but offers no view-scalable representation, so choosing VGGT is an enabling decision — the price of admission to the multi-view regime — not a matching-quality one. Second, learned capacity helps where the training data is dense and hurts outside it: LGv2 gains most at narrow baselines but falls 11 AUPRC behind the Sinkhorn baseline at Replica 135–180°, having internalised the moderate-viewpoint bias of ScanNet++ co-visible pairs; the un-learned baseline is thus the safe fallback exactly where learned heads extrapolate worst.

At 102 episodes, direct segment matchers and sparse keypoint aggregation are not statistically separated, so the offline regime split does not translate into a measurable closed-loop ordering. The supported gap is to RoMa’s dense pixel-vote baseline. Table 3 provides quantitative evidence on held-out real ScanNet++ scenes. Separately, Fig. 4 illustrates four-frame identity consistency during robot motion qualitatively, without ground-truth identities.

Limitations: (i) Training uses indoor ScanNet++ only; an outdoor source may close the VKITTI2 gap. (ii) Quantitative real-image evaluation is indoor and in-domain (ScanNet++ *val*, Table 3), not zero-shot; the in-house capture (Fig. 4)

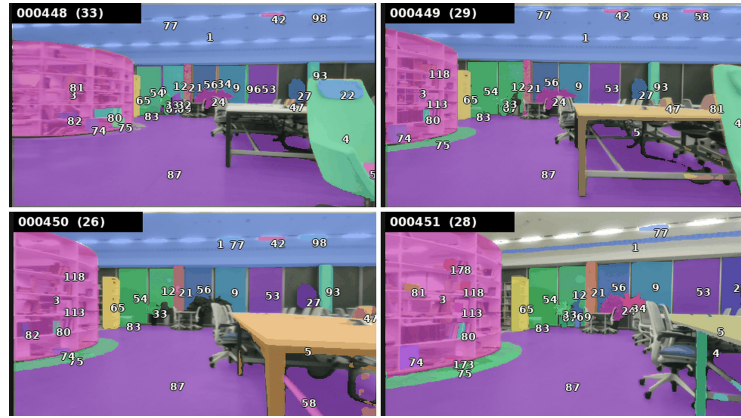


Fig. 4: Joint $N=4$ on real robot capture without retraining: views 448–451 matched in one joint pass. Colour encodes identity; headers give frame index and segment count. Only tracks spanning at least three of four views are drawn.

is qualitative, with no ground-truth identities. Real outdoor transfer remains untested. (iii) Matching quality is bounded by the supplied masks: a segmenter-sensitivity study across SAM granularity and FastSAM is left to future work. (iv) Closed-loop evaluation is confined to simulated indoor navigation.

5 Conclusion

We compared three segment-processing pipelines built on frozen geometry backbones. The result is regime-dependent rather than a single “best” model: LGv2 leads overall and at small-to-moderate baselines, including outdoor transfer, while Joint improves over pairwise DPT at the widest indoor baselines. The Sinkhorn baseline remains competitive in the extreme-rotation tail, consistent with learned heads’ sensitivity to the moderate-viewpoint training distribution. The backbone also constrains the available heads: MAST3R supports strong pairwise matching, while VGGT enables joint multi-view attention. Closed-loop evaluation supports no ordering among direct segment matchers and sparse keypoint aggregation, but separates all of them from the tested RoMa dense-vote baseline.

Future work should test outdoor training data and extreme-rotation sampling, routing between pairwise and joint heads using deployment cues, real outdoor zero-shot transfer, and sensitivity to segmenter choice and granularity.

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